



Introduction to Spring Security

What is Spring Security?

- Spring Security is a powerful and flexible security framework for Java-based applications.
- It provides **authentication**, **authorization**, and **session management** features.
- Spring Security is widely used in enterprise applications to protect against common security threats.
 - Cross-Site Scripting (XSS).
 - Cross-Site Request Forgery (CSRF).
 - Broken Authentication and Session Management.
 - Insufficient Cryptography.
 - Malware.
 - Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) Attacks.

Spring Security Architecture

- The Spring Security architecture consists of different modules that work together to provide a comprehensive security solution.
- The core module provides basic security features, such as authentication and authorization.
- Other modules extend the core module to support additional features, such as OAuth, OpenID Connect, and more.

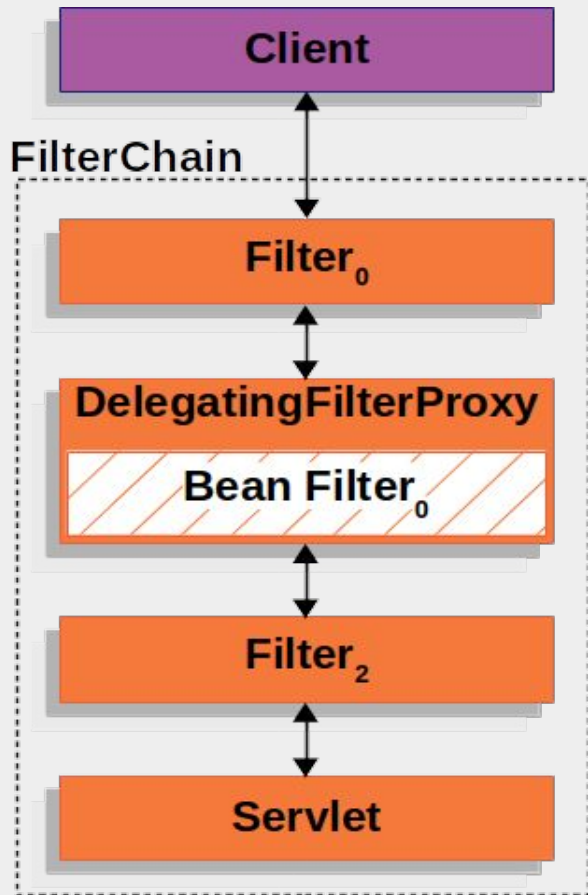
Authentication in Spring Security

- Authentication is the process of **verifying the identity of a user**.
- Spring Security supports multiple authentication mechanisms, such as **form-based login**, **HTTP basic authentication**, and **OAuth**.
- You can customize the authentication process using different authentication providers, such as **in-memory**, **JDBC**, and **LDAP**.

Authorization in Spring Security

- Authorization is the process of controlling access to resources based on user roles or permissions.
- Spring Security supports **role-based access control (RBAC)** and other types of authorization.
- You can define access control rules using **Spring Security's expression language**, which allows you to specify complex rules for controlling access to resources.

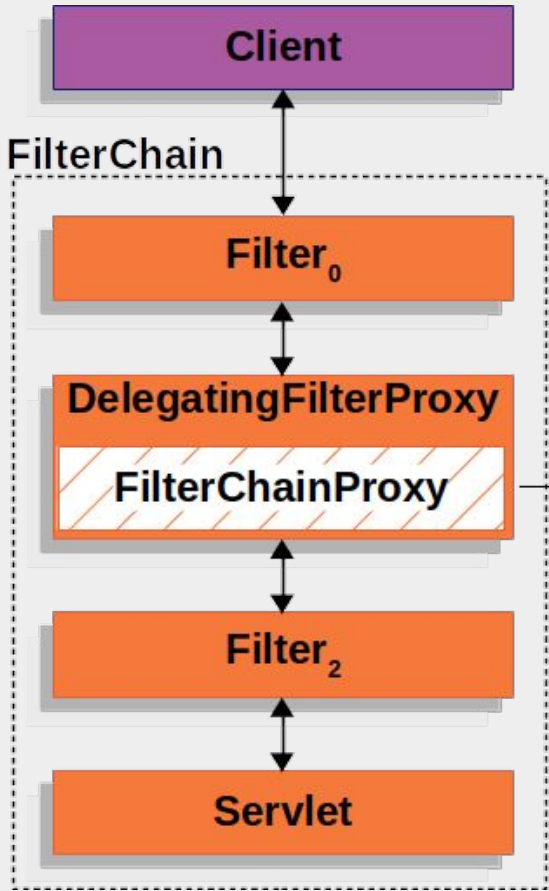
How Spring Security Works



DelegatingFilterProxy

Spring provides a Filter implementation named **DelegatingFilterProxy** that allows bridging between the Servlet container's lifecycle and Spring's **ApplicationContext**. The Servlet container allows registering Filter instances by using its own standards, but it is not aware of Spring-defined Beans. You can register **DelegatingFilterProxy** through the standard Servlet container mechanisms but delegate all the work to a Spring Bean that implements Filter.

How Spring Security Works

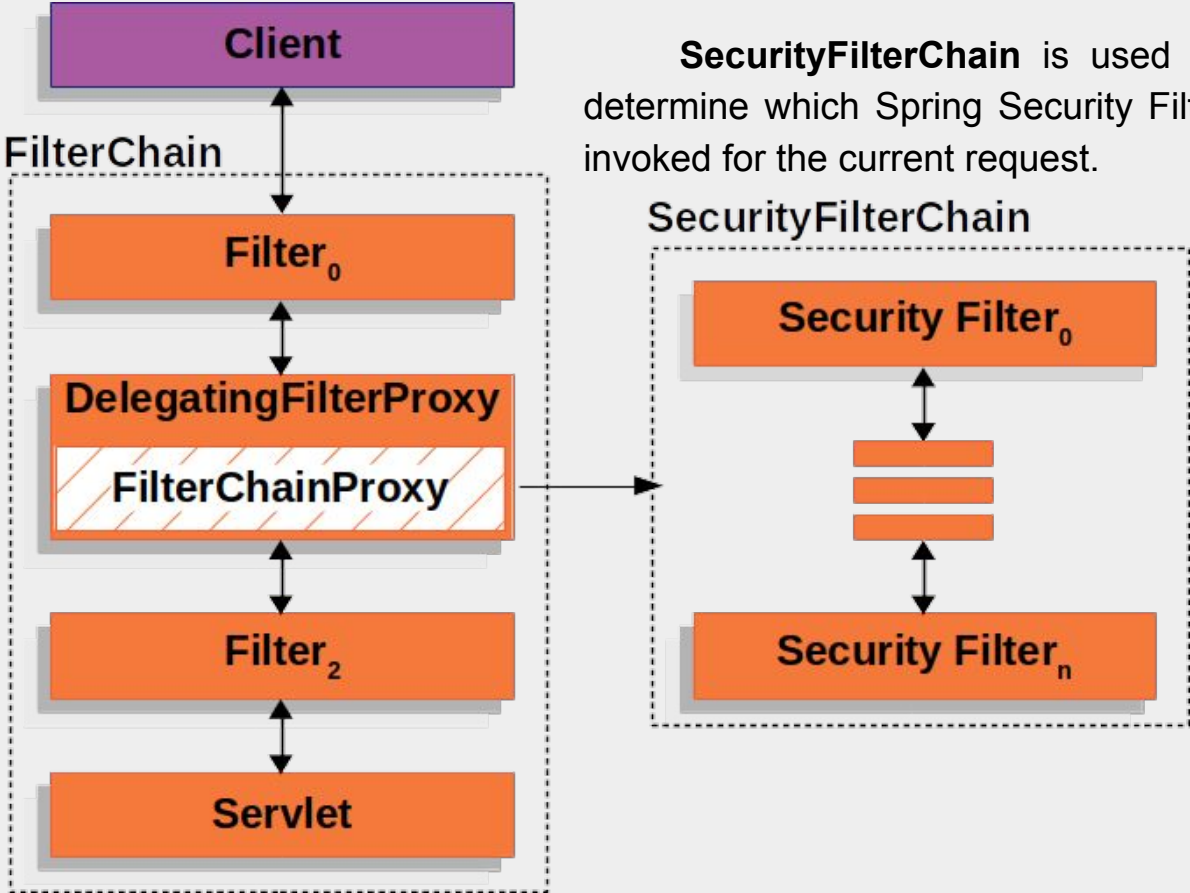


FilterChainProxy

Spring Security's Servlet support is contained within FilterChainProxy. FilterChainProxy is a special Filter provided by Spring Security that allows delegating to many Filter instances through **SecurityFilterChain**. Since FilterChainProxy is a Bean, it is typically wrapped in a DelegatingFilterProxy.



How Spring Security Works



SecurityFilterChain is used by **FilterChainProxy** to determine which Spring Security Filter instances should be invoked for the current request.

Spring Security Dependencies

```
<dependency>
  <groupId>org.springframework.security</groupId>
  <artifactId>spring-security-core</artifactId>
  <version>version</version>
</dependency>
<dependency>
  <groupId>org.springframework.security</groupId>
  <artifactId>spring-security-web</artifactId>
  <version>version</version>
</dependency>
<dependency>
  <groupId>org.springframework.security</groupId>
  <artifactId>spring-security-config</artifactId>
  <version>version</version>
</dependency>
```

Spring Security IN ACTION

Laurențiu Spilcă



Interview Questions